QR Code Generation

This test is conducted to validate that the QR code module correctly generates and parses certificate data. For the result, QR codes were successfully generated with structured payloads and parsed accurately during verification.

The system begins by decoding the QR code content. It first performs Base64 decoding, then applies XOR deobfuscation to reveal the original payload. The prefix **ECAS-CERT**: is validated, and the JSON structure is successfully parsed. The decrypted data contains the certificate ID, unique identifier, and the embedded Schnorr ZKP proof.

```
================================================================================
QR CODE PAYLOAD DECRYPTION - STARTED
================================================================================
[DEBUG] Encoded content length: 664 chars
[DEBUG] Base64 decode successful, obfuscated data length: 498 chars
[DEBUG] XOR deobfuscation complete, content preview: ECAS-CERT:eyJjZXJ0X2lkIjogIjEzIiwgInVuaXF1ZV9pZCI6...
[DEBUG] Prefix validation passed: 'ECAS-CERT:'
[DEBUG] JSON data decoded, length: 364 chars

[DEBUG] QR Payload Successfully Decrypted:
  - Keys found: ['cert_id', 'unique_id', 'zkp']
  - Data preview: {'cert_id': '13', 'unique_id': '690e811ba5f1527e71abbb60b4fbddec243407a1544d3946c4ed9bff12e596ec', 'zkp'
: '{"R": "M07HaMTbi27BMA4HGQhIKAiLn8uPy2bl40e6xuqyMEaKM9Egg0N2ze/ZfoPyA3C8/7KtxqfYaoOVyI4aNLmFiw...
  - ZKP proof: FOUND (length: 243 chars)
================================================================================
```

From the decrypted payload, the system identifies the ZKP proof components: the commitment **R** in **Base64**, the challenge **e** in hexadecimal, and the response **s** in hexadecimal. These values are essential for verifying the integrity of the certificate using zero-knowledge principles.

```
=== DECRYPTED QR DATA ===
Type: <class 'dict'>
Keys: ['cert_id', 'unique_id', 'zkp']
Full data: {'cert_id': '13', 'unique_id': '690e811ba5f1527e71abbb60b4fbddec243407a1544d3946c4ed9bff12e596ec', 'zkp': '{"R"
: "M07HaMTbi27BMA4HGQhIKAiLn8uPy2bl40e6xuqyMEaKM9Egg0N2ze/ZfoPyA3C8/7KtxqfYaoOVyI4aNLmFiw==", "e": "350a540c261ada4882f313
00afce2ad4d8ba5e4fe5bebc17067a7c4a488e6e51", "s": "35780c206bcf6ea832cacf791e610cb653f64ccbd0308675260b9973634b4d3f"}'}
=========================

ZKP PROOF VALUE: {"R": "M07HaMTbi27BMA4HGQhIKAiLn8uPy2bl40e6xuqyMEaKM9Egg0N2ze/ZfoPyA3C8/7KtxqfYaoOVyI4aNLmFiw==", "e": "3
50a540c261ada4882f31300afce2ad4d8ba5e4fe5bebc17067a7c4a488e6e51", "s": "35780c206bcf6ea832cacf791e610cb653f64ccbd030867526
0b9973634b4d3f"}
ZKP PROOF TYPE: <class 'str'>
Certificate data created: {"cert_id": "13", "cert_type": "Transcript of Records", "issue_date": "2025-12-03", "issuing_aut
hority": "Registrar Staff / St. Bridget College, Inc.", "student_id": "2013287", "student_name": "Zoro Roronoa", "timestam
p": "2025-12-03 12:05:48.100009+00:00", "unique_id": "690e811ba5f1527e71abbb60b4fbddec243407a1544d3946c4ed9bff12e596ec"}
ZKP proof from QR code provided - proceeding with ZKP verification
```

The verification routine decodes **R** into its 64-byte representation and converts **e** and **s** into large integers. Using the issuer's public key and the NIST P-256 curve, the system computes **gs** and **Ye**, then adds them to derive **R′**. This computed point is compared against the original commitment **R** from the QR code. The match confirms that the proof is valid, and the certificate data has not been altered.

```
================================================================================
QR ZKP VERIFICATION DEBUG - STARTED
================================================================================

[DEBUG] QR ZKP Proof Components:
  - R (base64): M07HaMTbi27BMA4HGQhIKAiLn8uPy2bl40e6xuqyMEaKM9Egg0...
  - e (hex): 350a540c261ada4882f31300afce2ad4d8ba5e4fe5bebc17067a7c4a488e6e51
  - s (hex): 35780c206bcf6ea832cacf791e610cb653f64ccbd0308675260b9973634b4d3f

[DEBUG] Decoded ZKP Components:
  - R bytes length: 64 bytes
  - e integer: 23990829519786306470931442472875519977340712807505927047728683228223291420241
  - s integer: 24184686317714727427282472073641365803742616032102876450651957954075658243391
```

```
[DEBUG] Certificate Data Hash: 5f6c7c061597437c88178971791526a6558d14ec4c14bcd50efec5df3cdc6e62

[DEBUG] Performing Schnorr ZKP Verification Calculations:
  - Using curve: NIST256p
  - Generator point G: (48439561293906451759052585252797914202762949526041747995844080717082404635286, 361342509567497957985851279195878819566111066729850150718771982535684144405109)
  - Computed g^s: (24518212933184555731657863209421009771740680196093814382355558230402562866199, 338993832157844761392972970704294085571312881963110909057306104765490672588830)
  - Computed Y^e: (22161994690929839251968475961059673548241830305195812474383666923651551087481, 550980413896089833789661616602692360391153945740671093939348704694783608588875)
  - Computed R' = g^s + Y^e: (23207145620888289091764104017538507885794347700889820933926966777401710882886, 625107256463228950942718242999381740902612578587178020270092640393945243662 19)
```

ZKP Proof Creation and Verification

The test was conducted to validate the creation of Schnorr-based ZKP proof and its verification during QR scanning. For the result, proof values (**R**, **e**, **s**) were generated and recomputed successfully, confirming privacy-preserving verification.

```
Starting QR code generation for certificate ID: 11
QR generation input data: {'cert_id': '11', 'unique_id': 'c03b7cdb2370e394e92054ffda800c3c94fee02580a404969bcfbbbac90609d7', 'zkp': '{"R": "E3fM11iZAAGLNF72DUWVM+62qr+G6fS+u+hxVAAqELznqZWB3WXzySy8kdwZUAFHKHnm/fQzt1Bk4Sg+iilzig==", "e": "7a00cadd42b7e558622da1be10516af7c3ae891710268a69dee72469ddd3d248", "s": "d6dea6daef115053bc0dcc5eaabdd773bda9ed8528c153219a35d03a26daa859"}'}
ZKP proof detected in certificate data
ZKP proof length: 243 chars
Encrypted data length: 664 chars
Successfully generated QR code for certificate ID: 11
```

```
[DEBUG] Performing Schnorr ZKP Verification Calculations:
  - Using curve: NIST256p
  - Generator point G: (48439561293906451759052585252797914202762949526041747995844080717082404635286, 3613425095674979579
8585127919587881956611106672985015071877198253568414405109)
  - Computed g^s: (35515468648816483816186728623426122122177024778922547709232493360616749092753, 344856191956265512872591
67415606531126813060640459427442927224224960362001406)
  - Computed Y^e: (53379929876459555411505206116838562455432263775217802674673761031199040687274, 3353890764648163987590031
30545864453381285784026850001733354483111386592206430)
  - Computed R' = g^s + Y^e: (99812865363398590335652433945266922950642842381426551794355421485591004219434, 4963836650164
49157703957836665568212052548115246875838248862764033335121542548)

[SUCCESS] QR ZKP VERIFICATION PASSED
  - The computed R point matches the commitment in the QR ZKP proof
```

ECC Keys and Digital Signature Creation

The test was conducted to ensure that ECC public-private key pairs are generated securely and to confirm that the system can sign certificate hashes using ECC and produce valid signatures. For the result keys were created and digital signatures were generated.

```
[ECC] Deterministic key pair created from seed ECAS-0OJOX..., fingerprint 0e886cdfd865ac1d...
[SIGN] Generating digital signature for cert #13 using priv_fp 0e886cdfd865ac1d... and pub_fp 225fc912abfbe39c...
[SIGN] Using private key fp 0e886cdfd865ac1d... and public key fp 225fc912abfbe39c...
[SIGN] Creating digital signature for payload hash 5f6c7c061597437c...
[SIGN] Digital signature created, length 96 chars
```

ECC Digital Signature Verification

This test is conducted to validate that the QR code module correctly generates and parses certificate data. For the result, QR codes were successfully generated with structured payloads and parsed accurately during verification.

```
ZKP verification result: True

Proceeding with digital signature verification
Stored signature (base64): MEYCIQDdTwEuFgYdu0Tx/EBpv/sWxo...
Public key: -----BEGIN PUBLIC KEY-----
MFk...
MFk...
Signature decoded, length: 72 bytes
Verifying signature against certificate data...
Signature verification result: VALID
Digital signature verification result: True
===== CERTIFICATE VERIFICATION COMPLETE =====
```

MFA OTP Generation and Validation

The test was conducted to ensure that MFA OTP codes are generated and validated correctly to secure access to verification. For the result, OTP codes were delivered via email and validated within the 5-minute window; incorrect codes were rejected.